USTECH
INNOVATIONS



INDUSTRY SOLUTIONS · CHEMICAL AND PHARMA

Process analytics for chemical and pharmaceutical manufacturing

Raw material verification, blend uniformity, drying endpoint and reaction monitoring with non-destructive NIR measurement.

Where measurement supports your process

Four points along the production flow where NIR results replace waiting for the laboratory.

01 Raw material identification

Incoming materials are checked non-destructively against classification models before release to production.

Parameters	Identity, grade, moisture
Instruments	TrueLine 17EC + caliX

02 Blend uniformity

Inline measurement on the blender shows when the blend has reached the defined homogeneity endpoint.

Parameters	Active content, excipient ratio, moisture
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + caliX

03 Granulation and drying endpoint

Moisture is tracked in real time and the endpoint signal is passed to the PLC.

Parameters	Moisture
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + ProChem

04 Reaction monitoring

Inline measurement in the process line follows conversion and endpoint without pulling samples.

Parameters	Solvent composition, conversion, moisture
Instruments	ProLine17EC · ProLine17ECS · ProLine2550 + ProChem

> The analyzers are not ATEX-certified and are not intended for classified hazardous (Ex) areas.

Instruments and software



TrueLine 17EC

Handheld NIR analyzer

- 908–1676 nm
- Measures at the sampling point
- Mobile app and cloud portal, Bluetooth
- Weight 1.4 kg



ProLine17EC

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- Typical analysis time 2–20 s
- 316 stainless steel, sapphire window, IP65
- Solid and liquid samples, no bypass



ProLine17ECS

Inline NIR analyzer

- 950–1650 nm, InGaAs photodiode array
- 2000 spectra per sample, S/N 23,000:1 (typical)
- 316 stainless steel, sapphire window, IP65
- Typical analysis time 2–20 s



ProLine2550

Inline FT-NIR analyzer

- 1,350–2,550 nm FT-NIR
- Flange mounting on flat chutes
- HTTP interface to the plant PLC (Ethernet, TCP/IP)
- IP65, -5 to +45 °C

ProChem

Instrument control and results. Shows measurements in real time, exports reports, and passes results to the plant PLC over an Ethernet-based HTTP interface.

caliX

Calibration workspace. Build, validate and transfer calibration models, with automated preprocessing and model selection.

Start with your application

Five answers are enough for us to tell you which instrument fits and where it should be installed.

1. What will be measured?
   Material and form: powders, granules, blends, solvents, reaction mixtures.
2. Which parameters do you need?
   For example identity, moisture, active content, solvent composition, conversion.
3. Where should the measurement take place?
   Receiving area, laboratory, next to the line, or directly on a blender, dryer or reactor line.
4. How do you measure these values today?
   Reference method and how long a result takes.
5. Where is the plant and what is your timeframe?
   Country and planned start.

Contact us

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